

Smart Traffic Signals: Vehicle Density Estimation & Dynamic Timing

Assigned to: Group 4

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Objective

The project aims to **estimate real-time vehicle density** from a camera feed or recorded traffic video and **dynamically adjust traffic signal durations**.

This helps **reduce average waiting time**, improve traffic throughput, and demonstrate an **AI-based traffic management system**.

How It Works

1. Video Input:

Accepts either a **live webcam feed** or **uploaded traffic video**.

2. Vehicle Detection:

Vehicles (cars, buses, trucks, bikes) are detected using **YOLOv8**, a real-time object detection model.

3. Lane Definition:

Users define **Regions of Interest (ROIs)** corresponding to traffic lanes for accurate counting.

4. Counting & Density Estimation:

Vehicles are counted per lane in each frame to estimate **lane-wise traffic density**.

5. Rule-Based Signal Logic:

Green signal duration is adjusted proportionally to lane density — **higher density = longer green time**.

6. Dashboard Features

- Upload video or use webcam feed
 - Live vehicle detection and counting
 - ROI (lane) management
 - Real-time metrics: FPS, vehicle counts, throughput
 - Optional robustness tests (dark/rain simulation)
 - Auto-saves ROI configuration for each video
 - Adaptive signal timing visualization
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Methodology

1. **Data Acquisition:**

- Video input from **webcam** or pre-recorded traffic footage.
- Optional ground-truth CSV to evaluate counting accuracy.

2. **Preprocessing:**

- Define **lane ROIs** manually or load from saved configuration.
- Optional simulation of **low-light or rain** conditions for robustness testing.

3. **Vehicle Detection:**

- YOLOv5 model detects vehicles in each frame.
- Only relevant vehicle classes (car, bus, truck, motorbike) are counted.

4. **Vehicle Tracking (Optional):**

- SORT tracker can be used to maintain unique IDs across frames.

5. **Lane-Wise Counting & Density Calculation:**

- Centroid of each detected vehicle is checked against lane ROIs.
- Counts per lane are used for traffic density estimation.

6. **Adaptive Signal Timing:**

- Lane green time is **proportional to vehicle density** within limits defined in the dashboard.
- Average green time is calculated dynamically for each frame.

7. **Visualization & Metrics:**

- Streamlit displays live video, lane counts, adaptive green times, and FPS metrics.
- Ground-truth comparison provides **MAE and MAPE** for evaluation.

 Model Used

Component	Model / Method	Purpose
Vehicle Detection	YOLOv5	Real-time detection of cars, buses, trucks, bikes
Control Logic	Rule-Based	Adaptive signal timing based on lane density
Interface	Streamlit	Live dashboard for visualization and monitoring

 Setup Instructions

1. Clone the repository

```
git clone https://github.com/aman-verse/Smart_Traffic_Signals.git
cd Smart_Traffic_Signals
```

2. Install dependencies
`pip install -r requirements.txt`
3. Run the dashboard
`streamlit run app.py`
4. Choose input
Upload traffic video (.mp4, .avi, .mov)
or Use webcam feed **for** live detection